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Course Code: CSA0903
Course Name: Programming in JAVA

Class activity

1. Write a program to store 15 integer values in an ArrayList and remove all duplicate elements without using another collection.

RemoveDuplicates.java	Output
<pre>1 import java.util.ArrayList; 2 import java.util.Scanner; 3 4 public class RemoveDuplicates { 5 public static void main(String[] args) { 6 Scanner sc = new Scanner(System.in); 7 ArrayList<Integer> list = new ArrayList<>(); 8 9 System.out.println("Enter size :"); 10 int n=sc.nextInt(); 11 System.out.println("Enter integers:"); 12 for (int i = 0; i < n; i++) { 13 list.add(sc.nextInt()); 14 } 15 16 for (int i = 0; i < list.size(); i++) { 17 for (int j = i + 1; j < list.size(); j++) { 18 if (list.get(i).equals(list.get(j))) { 19 list.remove(j); 20 j--; 21 } 22 } 23 } 24 25 System.out.println("After removing duplicates:"); 26 System.out.println(list); 27 } 28 }</pre>	Status: Accepted Memory Used: 12300 byte Execution Time: 0.177 sec Enter size : Enter integers: After removing duplicates: [1, 22, 23, 2, 0, 12, 3, 11] Input 12 122 23 2 0 12 22 0 3 2 3 11

2. Store employee IDs in a LinkedList and display them in both forward and reverse order.

EmployeeList.java	Output
<pre>1 import java.util.LinkedList; 2 import java.util.Scanner; 3 4 public class EmployeeList { 5 public static void main(String[] args) { 6 Scanner sc = new Scanner(System.in); 7 LinkedList<Integer> list = new LinkedList<>(); 8 9 // System.out.print("Enter number of employees: "); 10 int n = sc.nextInt(); 11 12 //System.out.println("Enter Employee IDs:"); 13 for (int i = 0; i < n; i++) { 14 list.add(sc.nextInt()); 15 } 16 17 System.out.println("Forward Order:"); 18 for (int id : list) 19 System.out.print(id + " "); 20 21 System.out.println("\nReverse Order:"); 22 for (int i = list.size() - 1; i >= 0; i--) 23 System.out.print(list.get(i) + " "); 24 } 25 }</pre>	Status: Accepted Memory Used: 12720 byte Execution Time: 0.203 sec Forward Order: 101 102 103 104 105 Reverse Order: 105 104 103 102 101 Input 5 101 102 103 104 105

3. Store email IDs entered by the user in a HashSet. Display only unique email IDs and count the total unique entries.

The screenshot shows a Java IDE with a file named `UniqueEmails.java`. The code imports `java.util.HashSet` and `java.util.Scanner`. It defines a `UniqueEmails` class with a `main` method. The `main` method prompts the user to enter the number of emails, then enters a loop to read email IDs and add them to a `HashSet`. After the loop, it prints the unique email IDs and the total count of unique emails.

```
1 import java.util.HashSet;
2 import java.util.Scanner;
3
4 public class UniqueEmails {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7         HashSet<String> set = new HashSet<>();
8
9         System.out.print("Enter number of emails: ");
10        int n = sc.nextInt();
11        sc.nextLine();
12
13        System.out.println("Enter Email IDs:");
14        for (int i = 0; i < n; i++) {
15            set.add(sc.nextLine());
16        }
17
18        System.out.println("Unique Email IDs:");
19        for (String email : set)
20            System.out.println(email);
21
22        System.out.println("Total Unique Emails: " + set.size());
23    }
24 }
```

The output shows the program's execution. It prompts for the number of emails (5) and then for the email IDs. The unique email IDs are displayed, and the total count is 4.

Status: **Accepted**
Memory Used: 12640 byte | Execution Time: 0.282 sec

Enter number of emails: Enter Email IDs:
Unique Email IDs:
test@gmail.com
xyz@gmail.com
abc@gmail.com
Total Unique Emails: 4

Input
5
abc@gmail.com
xyz@gmail.com
abc@gmail.com
test@gmail.com
xyz@gmail.com

4. Write a Java program to count the frequency of each word in a paragraph using a HashMap.

The screenshot shows a Java IDE with a file named `WordFrequency.java`. The code imports `java.util.HashMap` and `java.util.Scanner`. It defines a `WordFrequency` class with a `main` method. The `main` method prompts the user to enter a paragraph, splits it into words, and uses a `HashMap` to count the frequency of each word. It then prints the word frequencies.

```
1 import java.util.HashMap;
2 import java.util.Scanner;
3
4 public class WordFrequency {
5     public static void main(String[] args) {
6         Scanner sc = new Scanner(System.in);
7
8         System.out.println("Enter a paragraph:");
9         String text = sc.nextLine();
10
11        String[] words = text.split(" ");
12
13        HashMap<String, Integer> map = new HashMap<>();
14
15        for (String word : words) {
16            if (map.containsKey(word))
17                map.put(word, map.get(word) + 1);
18            else
19                map.put(word, 1);
20        }
21
22        System.out.println("Word Frequencies:");
23        for (String word : map.keySet()) {
24            System.out.println(word + " : " + map.get(word));
25        }
26    }
27 }
```

The output shows the program's execution. It prompts for a paragraph and then displays the word frequencies.

Status: **Accepted**
Memory Used: 12956 byte | Execution Time: 0.288 sec

Enter a paragraph:
Word Frequencies:
java : 2
powerful : 1
is : 2
easy : 1

Input
java is easy java is powerful